

# Module 4

## Circuit Analysis in the S Domain

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# Laplace Transform

Laplace transformation is a powerful method, which involves turning differential equations into algebraic equations, thus greatly facilitating the solution process.

**We use the Laplace transformation to transform the circuit from the time domain to the frequency domain**, obtain the solution, and apply the inverse Laplace transform to the result to transform it back to the time domain. This is similar to the process we adopted while working with phasors.

**The Laplace transform is significant because:**

1. It can be applied to a wider variety of inputs than phasor analysis.
2. It provides an easy way to solve circuit problems involving initial conditions, because it allows us to work with algebraic equations instead of differential equations.
3. It is capable of providing us, in one single operation, the total response of the circuit comprising both the natural and forced responses.

# Laplace Transform

Given a function  $f(t)$ , its Laplace transform is denoted by  $F(s)$  or by  $L\{f(t)\}$  and is defined by

$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st} dt$$

This integral is a definite integral with respect to time. Hence, the result of integration is independent of time and only involves the variable “s.”

where ‘s’ is a complex variable given by  $s = \sigma + j\omega$

The argument  $st$  of the exponent  $e$  in definition must be dimensionless. Therefore it follows that  $s$  has the dimensions of frequency and units of inverse seconds ( $s^{-1}$ ) or frequency.

The lower limit is specified as  $0^-$  to indicate time just before  $t = 0$ .

We use  $t = 0^-$  as the lower limit to include the origin and capture any discontinuity of  $f(t)$  at  $t = 0$ ; this will accommodate functions—such as singularity functions that may be discontinuous at  $t=0$ .

# Laplace Transform

Therefore, the Laplace transform is an integral transformation of a function  $f(t)$  from the time domain into the complex frequency domain, giving  $F(s)$ .

The differential equations which represent the circuit in the time domain, can be represented as algebraic equations by applying their Laplace transform, which corresponds to  $F(s)$  representing the circuit in the frequency domain.

In this definition  $f(t)$  for  $t < 0$  is ignored. To ensure this  $f(t)$  is multiplied by a unit step and is written as  $f(t)u(t)$ , for all  $t \geq 0$ .

**This known as the one-sided (or unilateral) Laplace transform.**

The two-sided (or bilateral) Laplace transform is given by

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st} dt$$

The one-sided Laplace transform is adequate for our purposes now.

# Laplace Transform

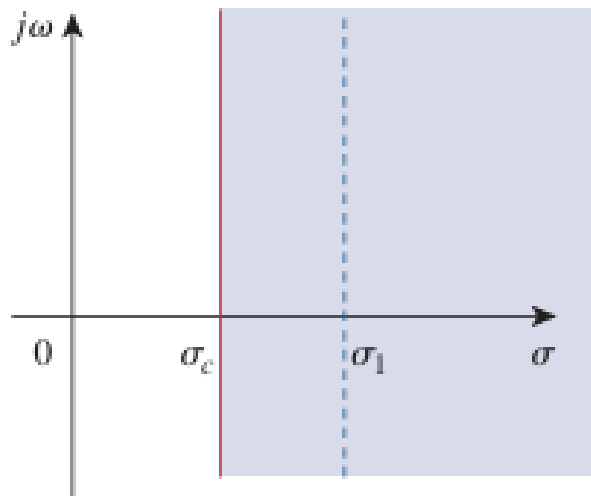
In order for  $f(t)$  to have a Laplace transform, the integral in the definition **must converge to a finite value**.

Since  $|e^{j\omega t}| = 1$  for any value of  $t$ , the integral converges when,

$$\int_{0^-}^{\infty} e^{-\sigma t} |f(t)| dt < \infty \quad \text{for some real value } \sigma = \sigma_c$$

This is the **convergence criteria**.

Thus, the **region of convergence** for the Laplace transform is  $\text{Re}(s) = \sigma > \sigma_c$



**In this region,  $|F(s)| < \infty$  and  $F(s)$  exists.**  $F(s)$  is undefined outside the region of convergence.

# Laplace Transform

For the transform to exist (i.e., for the integral to converge), the following condition must be met:

$$\int_{-\infty}^{\infty} |x(t)e^{-st}| dt < \infty.$$

**Role of  $\sigma$ :**  $s = \sigma + j\omega$

$$e^{-st} = e^{-(\sigma + j\omega)t} = e^{-\sigma t} + e^{-j\omega t}$$

If  $\sigma$  is positive, the term  $e^{-\sigma t}$  acts as an exponential decay factor  $t \geq 0$ . As  $t$  approaches *infinity*, the exponential term approaches *zero*, attenuating the signal  $x(t)$ .

For right sided signal, to ensure convergence for  $t \geq 0$ , a positive value of  $\sigma$  is required thereby preventing the integral from diverging.

# Laplace Transform

Inverse Laplace transform is given by

$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j} \int_{\sigma_1 - j\infty}^{\sigma_1 + j\infty} F(s)e^{st} ds$$

where the integration is performed along a straight line (  $\sigma_1 + j\omega$ ,  $-\infty < \omega < \infty$  ) in the region of convergence,  $\sigma_1 > \sigma_c$  .

The functions  $f(t)$  and  $F(s)$  are regarded as a Laplace transform pair where,

$$f(t) \quad \Leftrightarrow \quad F(s)$$

# Laplace Transform

1. Determine the Laplace transform of each of the following functions: (a)  $u(t)$ , (b)  $e^{-at} u(t)$ ,  $a \geq 0$ , and (c)  $\delta(t)$ .

$$\begin{aligned} \text{(a) } u(t) \quad \mathcal{L}[u(t)] &= \int_{0^-}^{\infty} 1 e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} \\ &= -\frac{1}{s}(0) + \frac{1}{s}(1) = \frac{1}{s} \end{aligned}$$

$$\begin{aligned} \text{(b) } e^{-at} u(t), a \geq 0 \quad \mathcal{L}[e^{-at} u(t)] &= \int_{0^-}^{\infty} e^{-at} e^{-st} dt \\ &= -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^{\infty} = \frac{1}{s+a} \end{aligned}$$

$$\text{(c) } \delta(t) \quad \mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t) e^{-st} dt = e^{-0} = 1$$

# Laplace Transform

2. Find the Laplace transforms of these functions: (a) the ramp function  $r(t) = t u(t)$ ; (b)  $A e^{-at} u(t)$  and  $B e^{-j\omega t} u(t)$ .

Answer:  $1/s^2$ ,  $A / (s + a)$ ,  $B / (s + j\omega)$ .

3. Determine the Laplace transform of  $f(t) = \sin \omega t u(t)$ .

$$\begin{aligned} F(s) = \mathcal{L}[\sin \omega t] &= \int_0^{\infty} (\sin \omega t) e^{-st} dt = \int_0^{\infty} \left( \frac{e^{j\omega t} - e^{-j\omega t}}{2j} \right) e^{-st} dt \\ &= \frac{1}{2j} \int_0^{\infty} (e^{-(s-j\omega)t} - e^{-(s+j\omega)t}) dt \\ &= \frac{1}{2j} \left( \frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right) = \frac{\omega}{s^2 + \omega^2} \end{aligned}$$

4. Find the Laplace transform of  $f(t) = 50 \cos \omega t u(t)$ .

Answer:  $50s / (s^2 + \omega^2)$

Next class: Laplace Transform - Properties